

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.025
##
## $beta.var.scale
## [1] 0.8
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.3975  0.4325  0.4417  0.4422  0.4517  0.4883

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09667 0.14417 0.14029 0.17750 0.82917
```

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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1000 0.1483 0.1438 0.1817 0.7325
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102021 0.031667 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.077472 0.024375 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005833 0.015000 0.162479 0.119167 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.1883 0.2200 0.2292 0.2296 0.2392 0.2800

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0550 0.1067 0.1104 0.1483 0.8517
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.05833	0.11083	0.11364	0.15333	0.70833
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000000	0.0008333	0.0466667	0.4090410	1.0000000	1.0000000
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000000	0.0008333	0.0041667	0.2431021	0.3197917	1.0000000
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.1033	1.0000	0.6919	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.05583	0.07250	0.08083	0.08204	0.08917	0.17833

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.03500	0.08333	0.09023	0.12750	0.85167
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.03667	0.08833	0.09429	0.13500	0.63250
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.4067	0.5842	0.6230	1.0000	1.0000
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.0000	0.0025	0.2651	0.5475	1.0000
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.5667	1.0000	0.7828	1.0000	1.0000

```

--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02000 0.04583 0.05333 0.05504 0.06167 0.15583

```

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beta.ar (Location: fit[[4]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.06000 0.07304 0.10583 0.83417
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar (Location: fit[[4]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02000 0.06500 0.07689 0.11333 0.63833
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3875  0.7642  0.6494  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9567  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2582  0.5971  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  1.0000  1.0000  0.8642  1.0000  1.0000

```

```

--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01583 0.03750 0.04583 0.04729 0.05500 0.13917

```

```

beta.ar (Location: fit[[5]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.05500 0.06800 0.09750 0.78500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02000 0.06000 0.07226 0.10500 0.68750
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01417 0.51917 0.51034 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.8317  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2390  0.4475  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.8425  1.0000  0.8301  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.008333 0.020833 0.025833 0.027653 0.032500 0.077500

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01083 0.04333 0.05845 0.08500 0.80750
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.04833 0.06285 0.09333 0.54083
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.2750  0.4407  1.0000  1.0000
  Modality 3 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7350	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2194	0.3106	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7017	1.0000	0.7799	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01417	0.03083	0.04167	0.04422	0.05333	0.11250

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.009167	0.034167	0.051069	0.073333	0.734167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03833	0.05570	0.08167	0.57000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2809	0.6608	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9992	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1285	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0200	0.7242	0.5806	0.9892	1.0000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1508	0.1792	0.1950	0.1976	0.2108	0.3967

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004167 0.027500 0.062332 0.072500 0.751667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1

```

gamma.ar (Location: fit[[8]]\$gamma.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00500 0.03167 0.05864 0.08000 0.53250
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1656 0.3150 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.6758 0.7692 0.7688 0.8725 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.03251 0.00000 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1387 0.2325 1.0000

```

--- Summary for K = 9 ---

Z.ar (Location: fit[[9]]\$Z.ar):

```

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.7100 0.8242 0.8350 0.8344 0.8458 0.8733

```

beta.ar (Location: fit[[9]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.013333 0.051129 0.040833 0.866667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1

```

```

gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.015000 0.038507 0.043333 0.472500
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.01117 0.000000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 6.87e-05 0.00e+00 1.00e+00
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.00435 0.000000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.01166 0.000000 1.00000

```